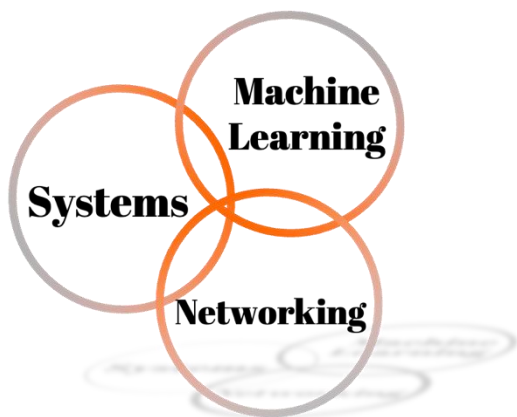
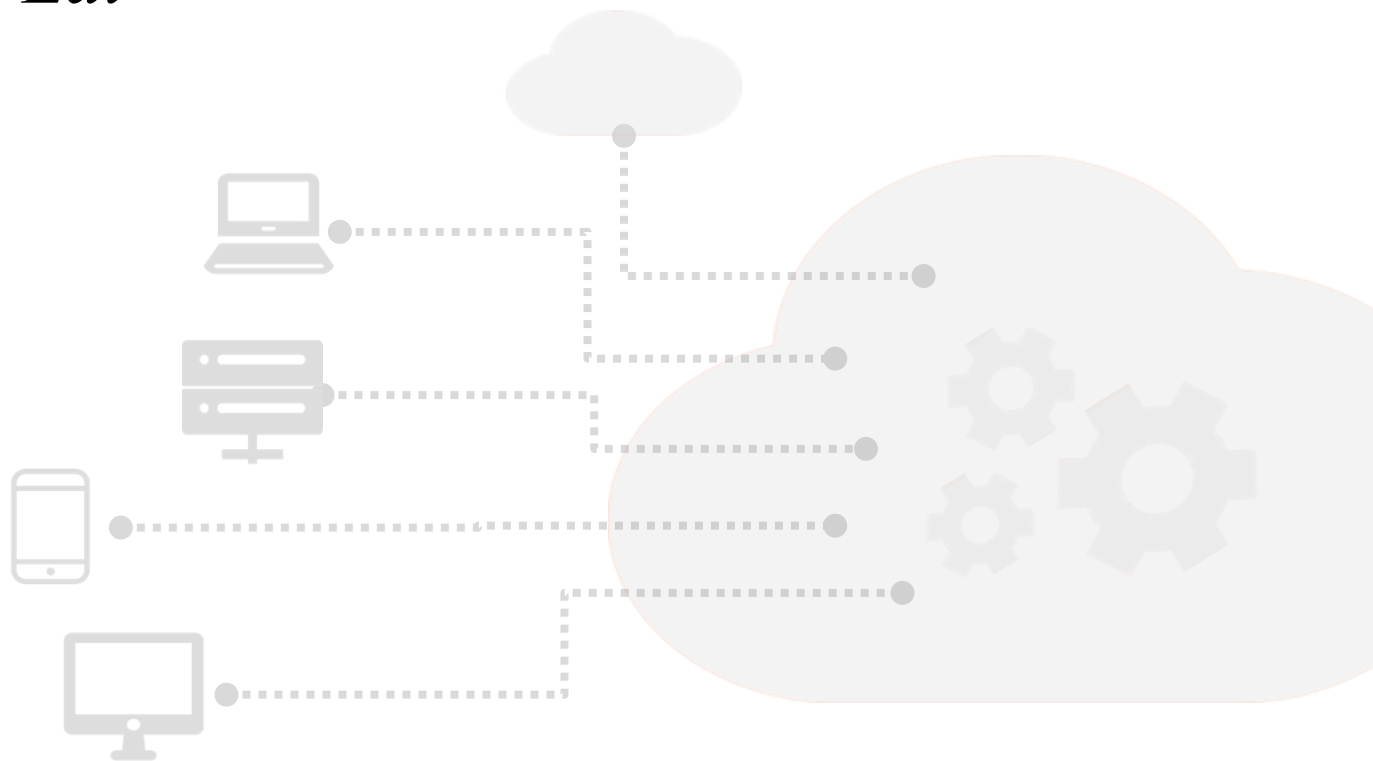




CS 598: Systems for GenAI

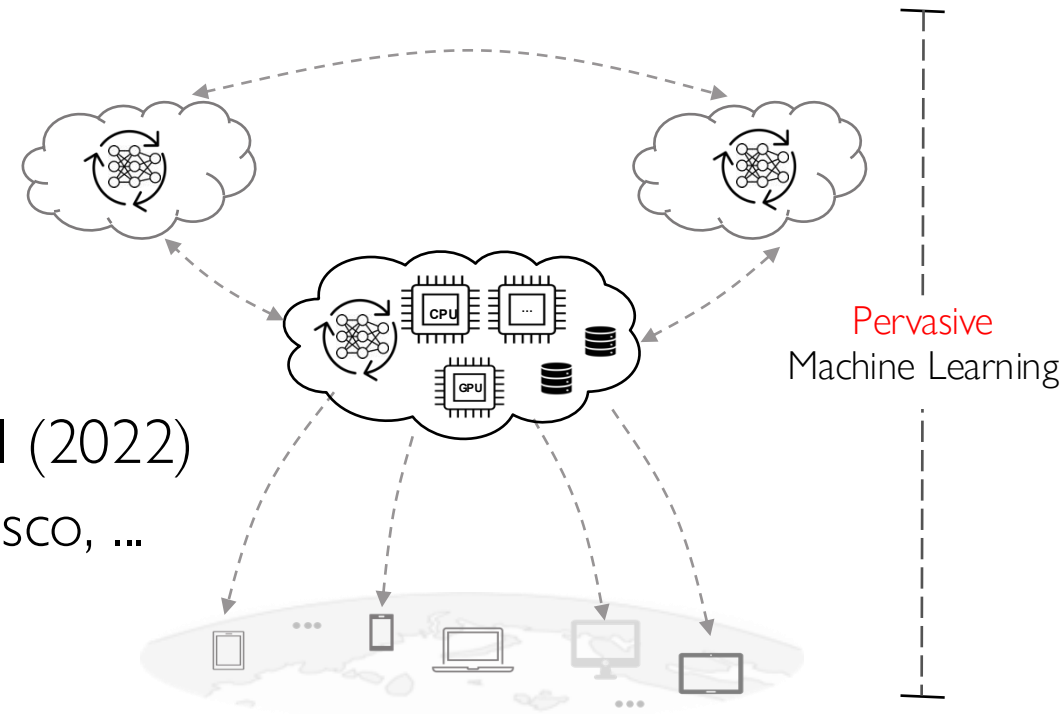
Instructor: Fan Lai

I ILLINOIS



About Instructor (Fan)

- **Assistant Professor in CS**
 - PhD'23: "Minimalist Systems for Pervasive Machine Learning"
- **Research interest**
 - Cloud ML (LLMs, Image/Video Gen)
 - On-device ML, ML4Sys, ML efficiency algs
- **Industry**
 - Visiting Faculty@Google (2023-2024), Meta AI (2022)
 - MLSys adoptions at Google, Meta, LinkedIn, Cisco, ...
- **Office hours:** SC 3128, Friday 2–3 PM



About Our Two TAs (Jimmy and Yuhan)



Jimmy Shong

(Research: Cloud ML training & inference)



Yuhan Ding

(Research: Distributed training & inference)

Office Hours: Wed 7-8 PM ([Zoom](#), or in-person upon appointment)

Status and Prerequisites

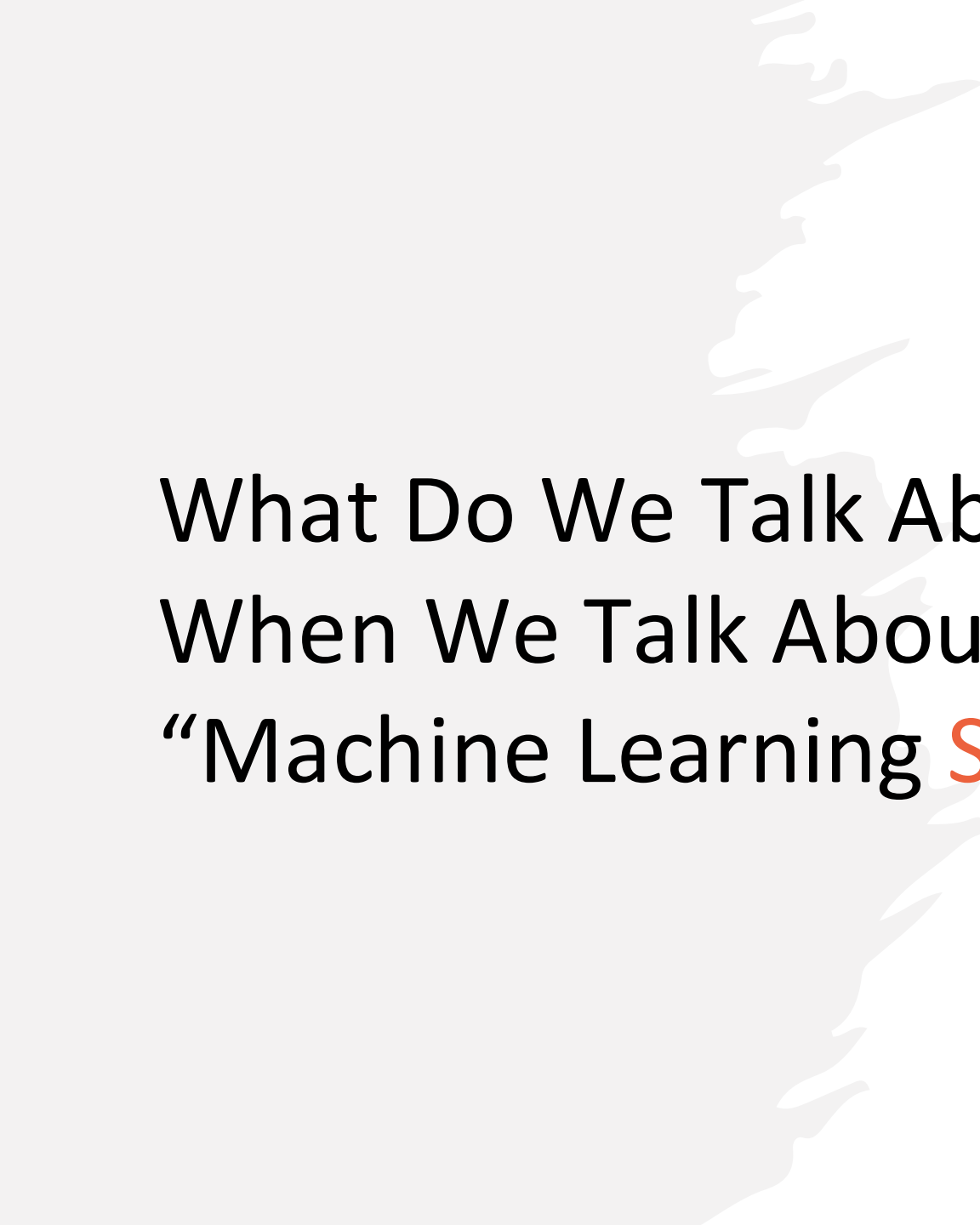
- As of today: ~93 registered
 - If you are not planning to take the class, please drop ASAP
- Systems Courses: OS/databases/distributed systems/networking
 - Equivalent courses are acceptable as well
- ML/AI course is helpful but not required
- Good programming skills (e.g., PyTorch, TensorFlow, ...)
 - Build systems for course project

Course Schedule

- Webpage: <https://github.com/fanlai0990/cs598>
- Discussion, Questions: Please Use [Canvas](#)
- In-person lectures, discussion
- Feel free to drop us email too :)

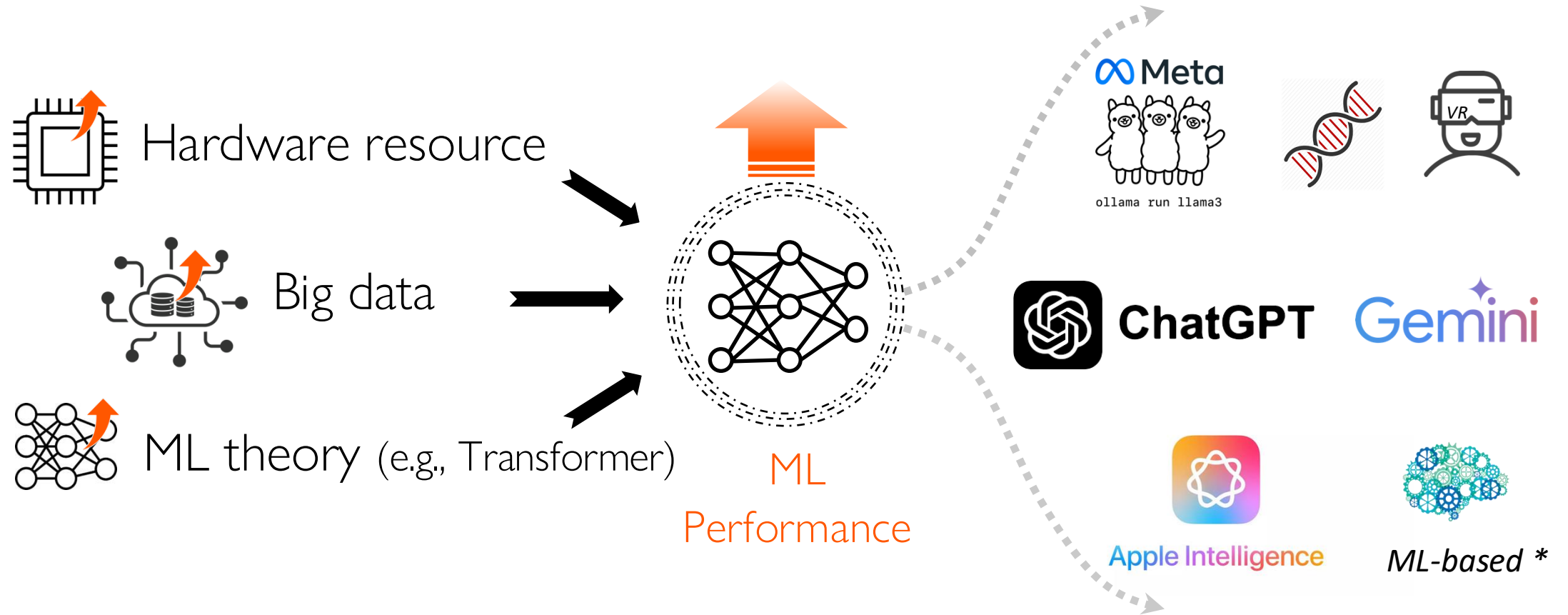
Agenda

- What will you learn in this course?
 - GenAI, systems, networks, ...
- What will you do in this course?
 - Participation, presentation, paper review, course project, ...



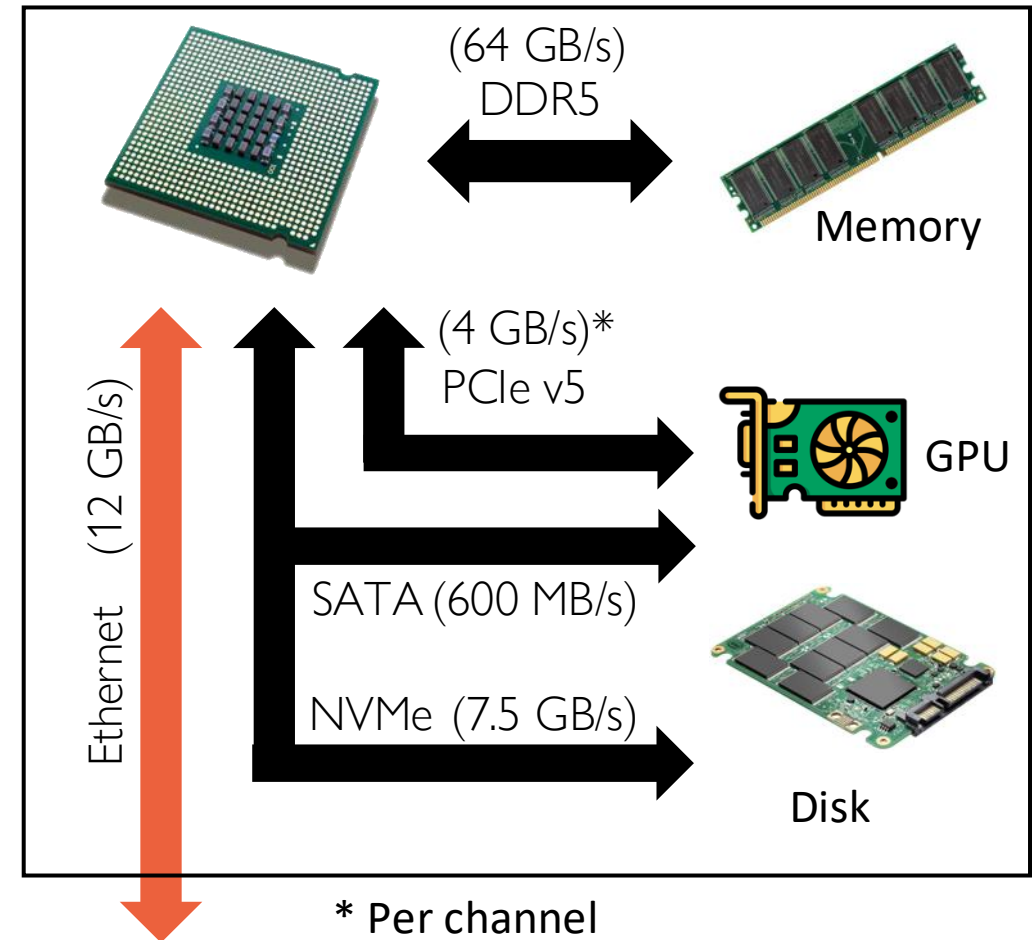
What Do We Talk About When We Talk About “Machine Learning **Systems**”

Cloud Enabled Last Leap in Machine Learning (ML)

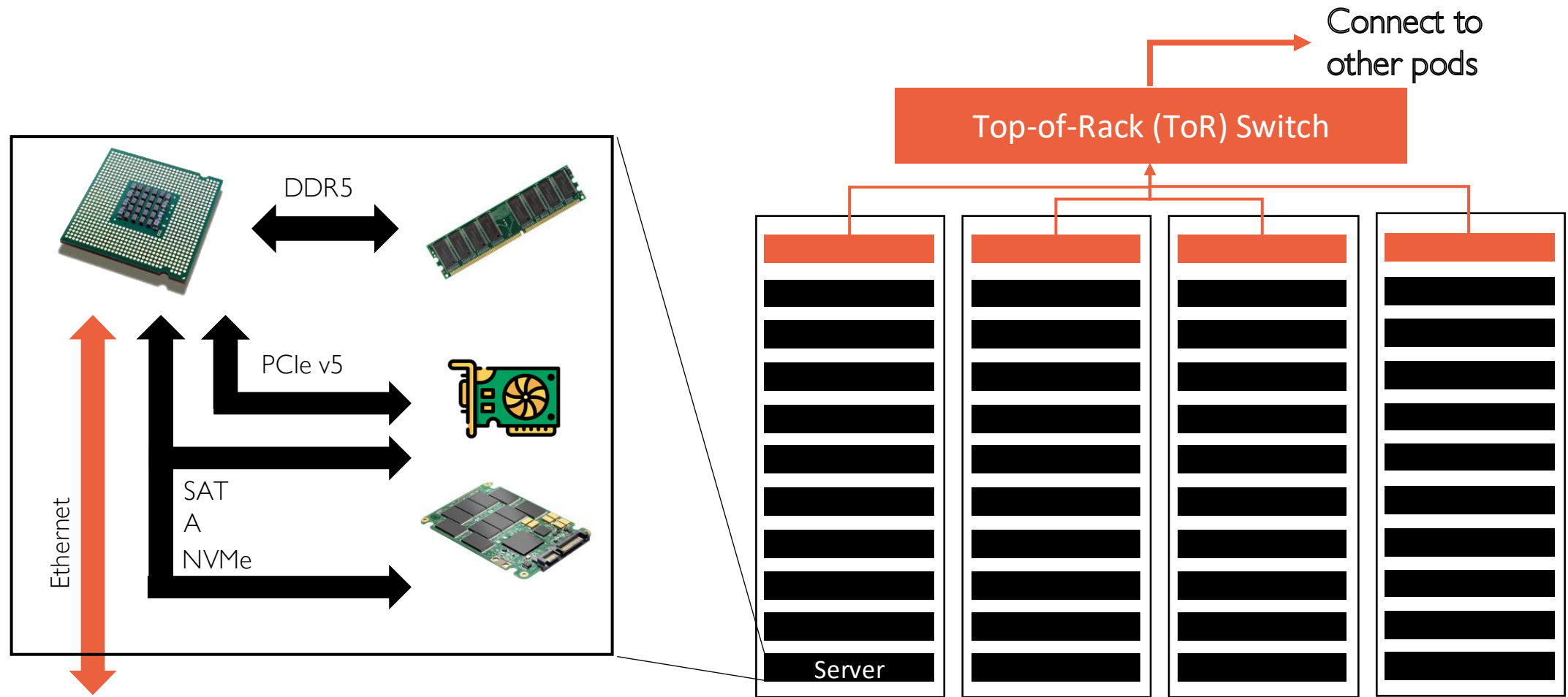


What's in a (Simplified) AI Server?

- **Interconnected compute and storage resources**
 - Different bandwidth and latency constraints
- **Simplified diagram**
 - Doesn't include faster networks such as RDMA, dedicated GPU interconnects such as NVLink

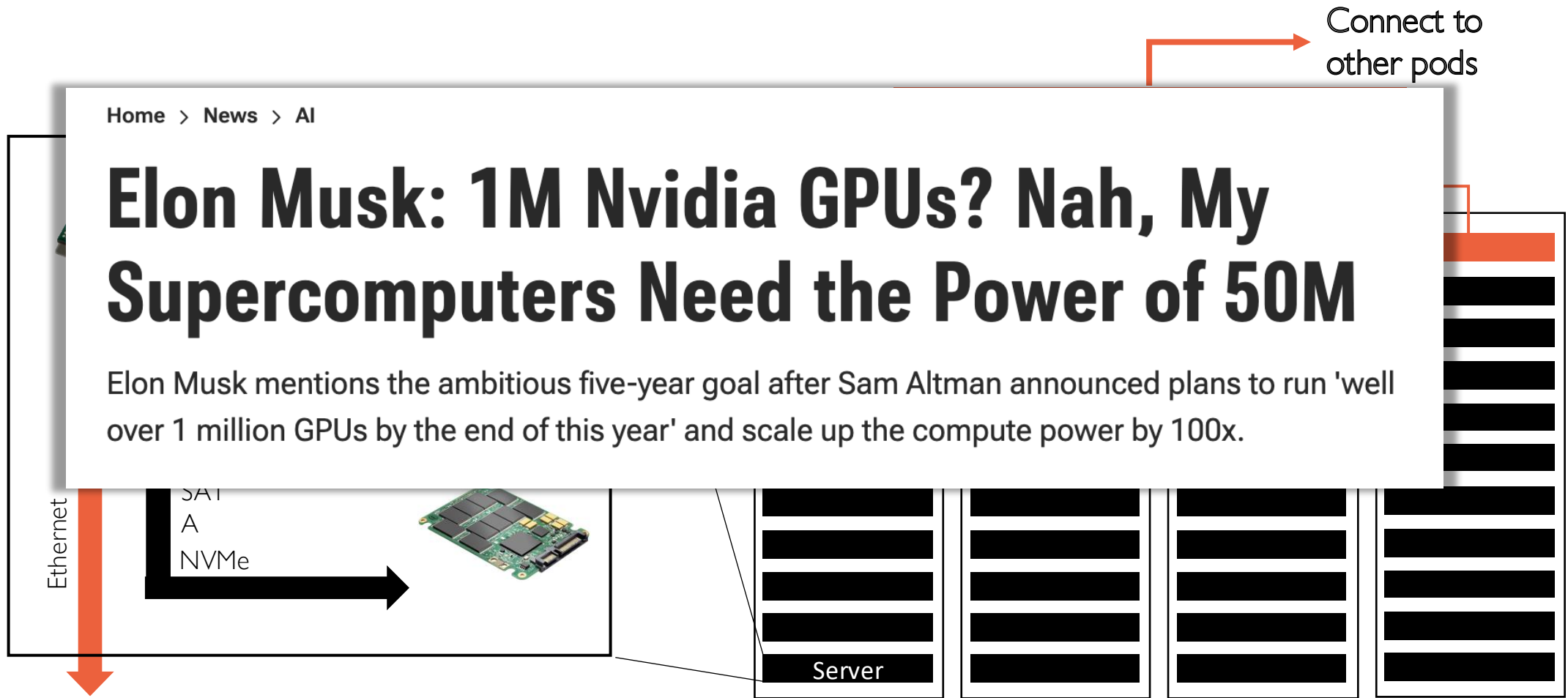


Cloud Datacenter: Scale-Out Servers



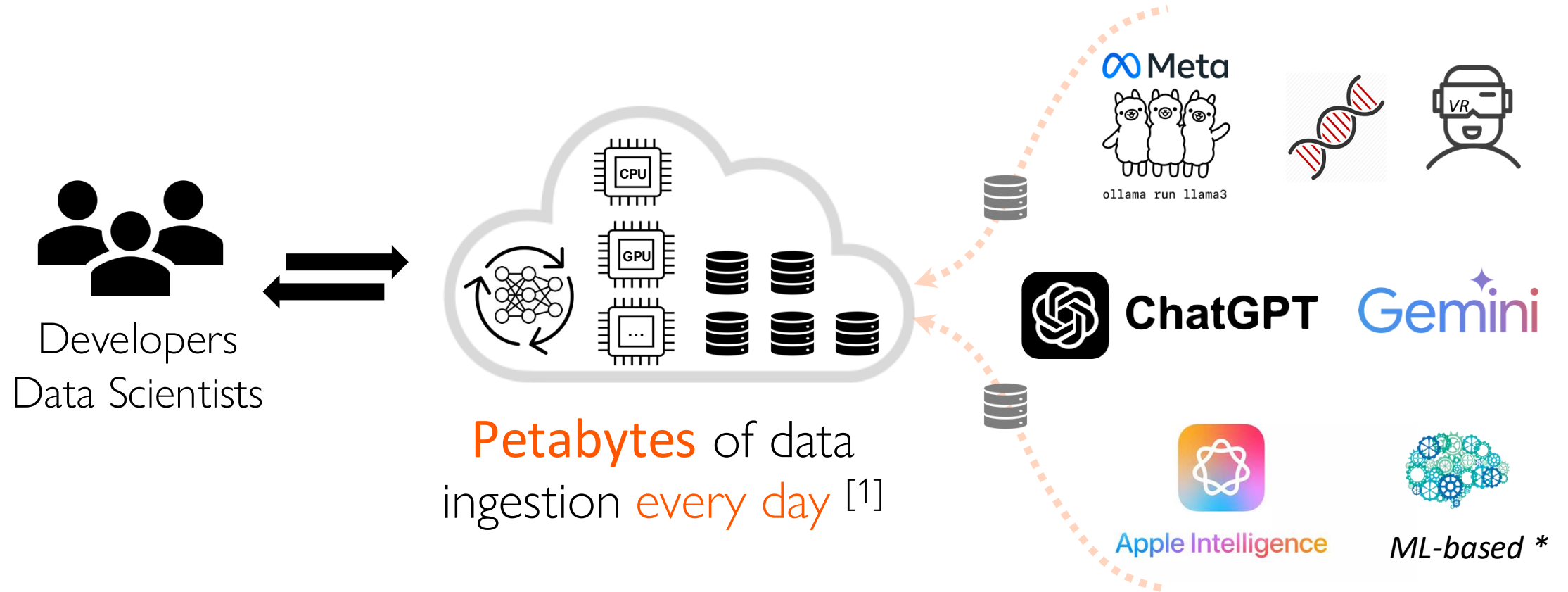
Meta's AI datacenter houses 16K of its 24K GPUs for training the Llama 3 model

Cloud Datacenter: Scale-Out Servers



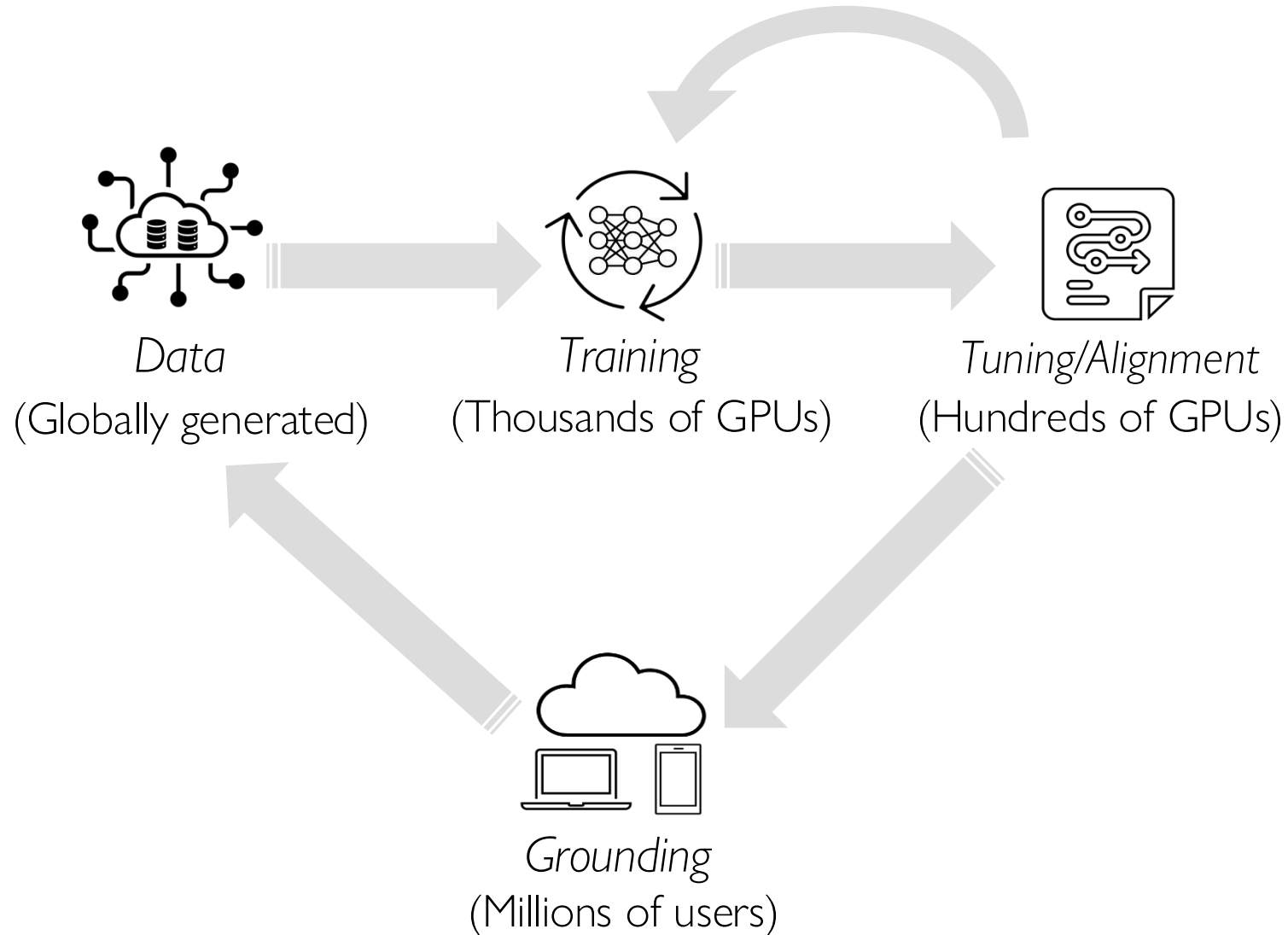
Meta's AI datacenter houses 16K of its 24K GPUs for training the Llama 3 model

Cloud Faces Skyrocketing ML Workloads

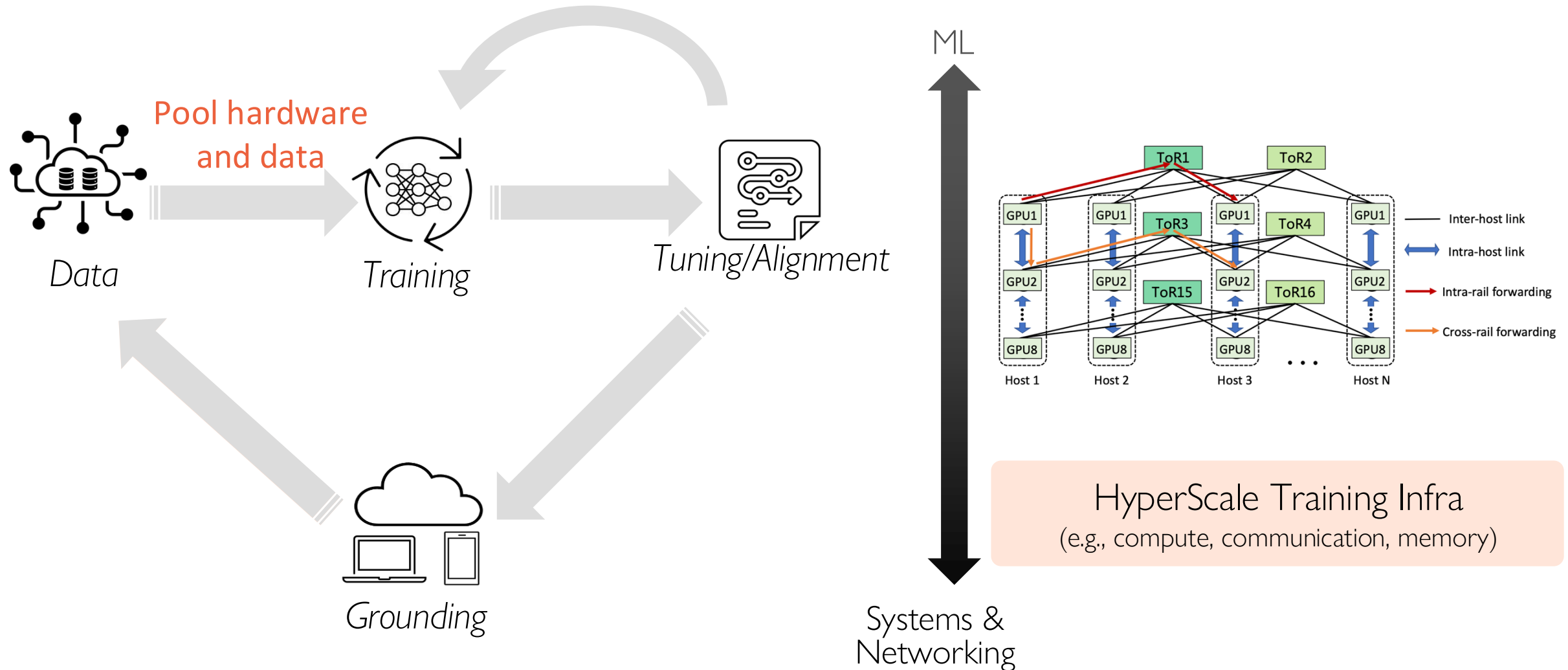


[1] "Sustainable AI: Environment Implications, Challenges and Opportunities", Meta, MLSys'22

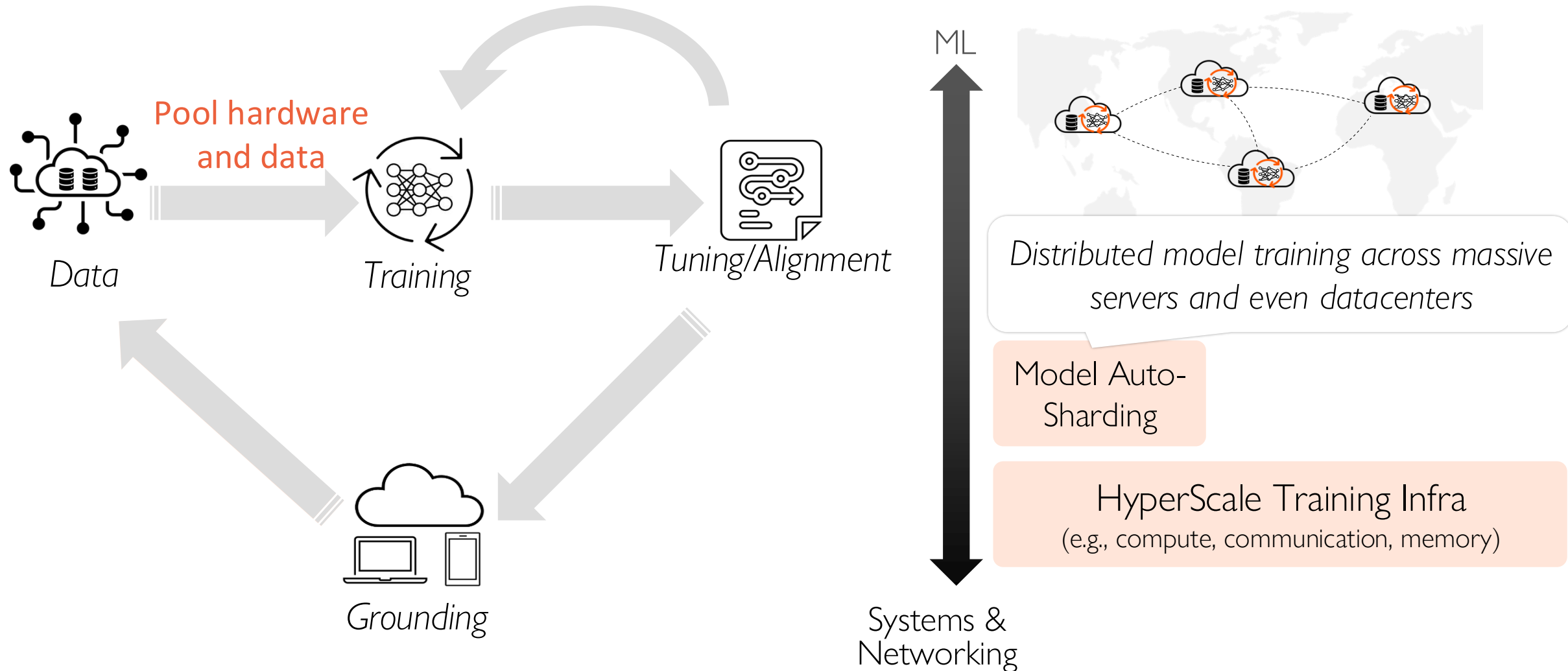
GenAI Lifecycle Demands Great Systems Support



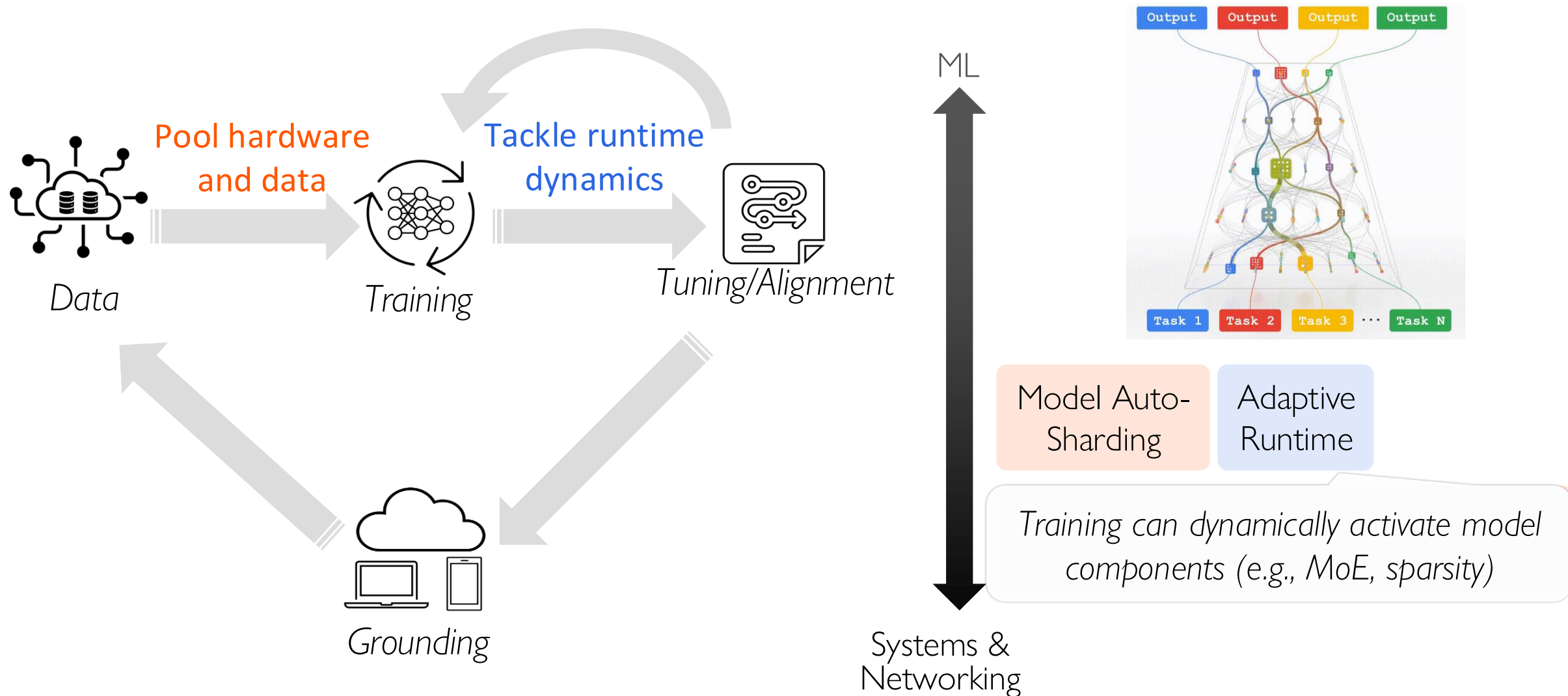
GenAI Lifecycle Demands Great Systems Support



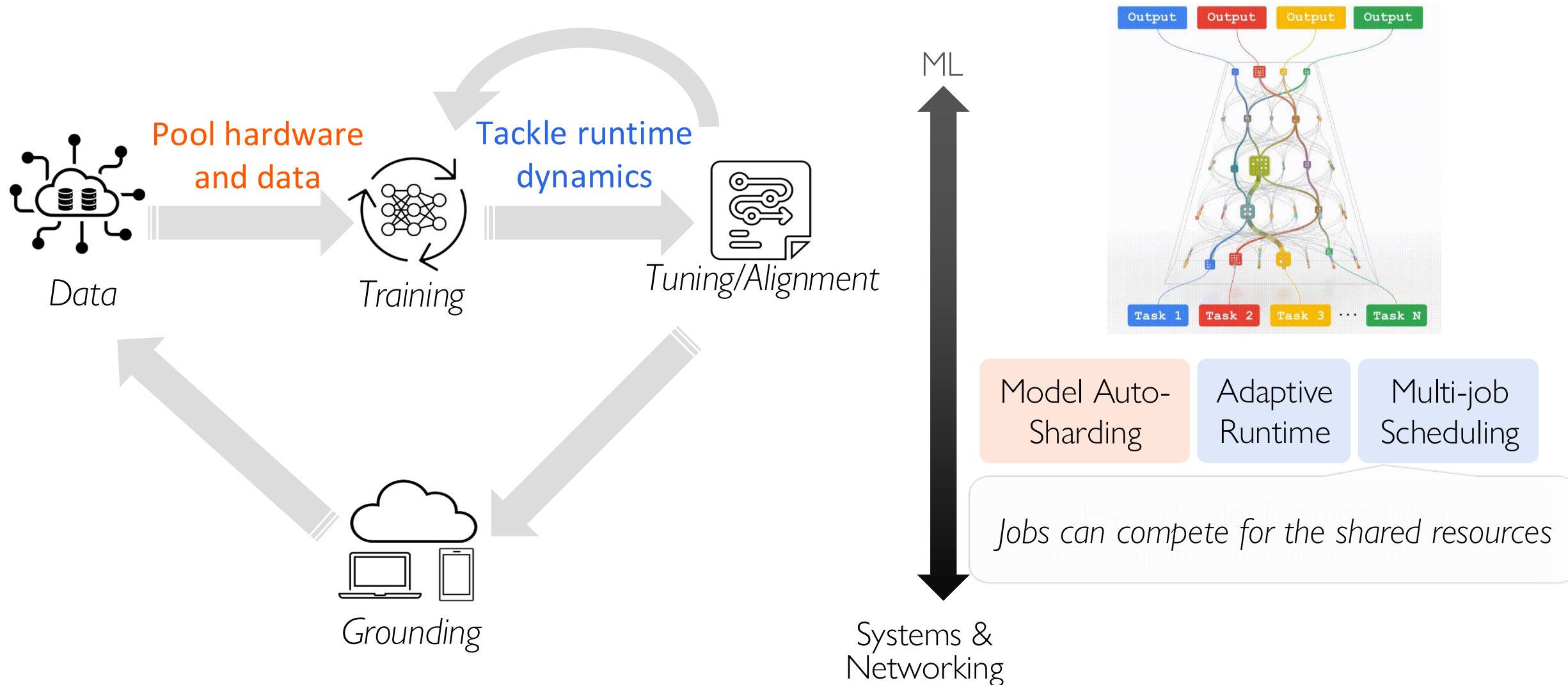
GenAI Lifecycle Demands Great Systems Support



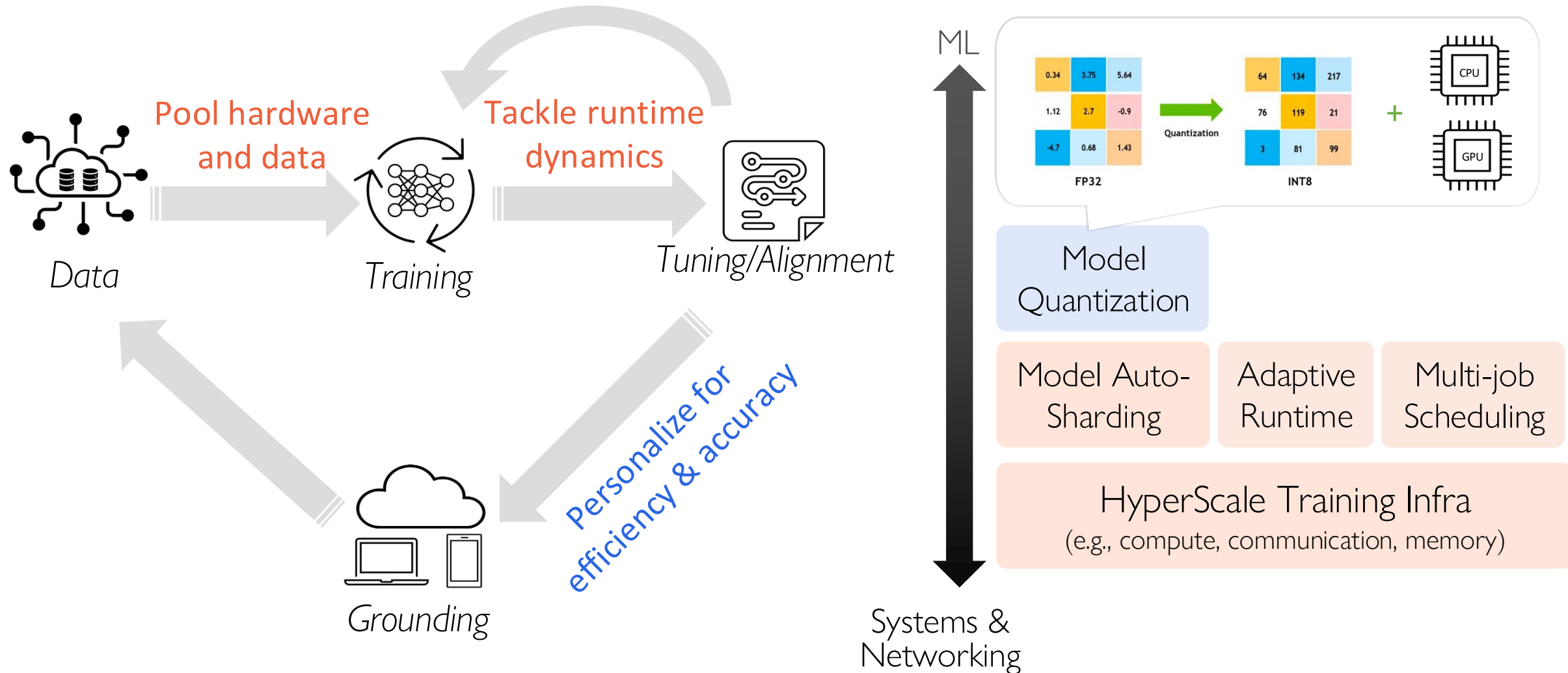
GenAI Lifecycle Demands Great Systems Support



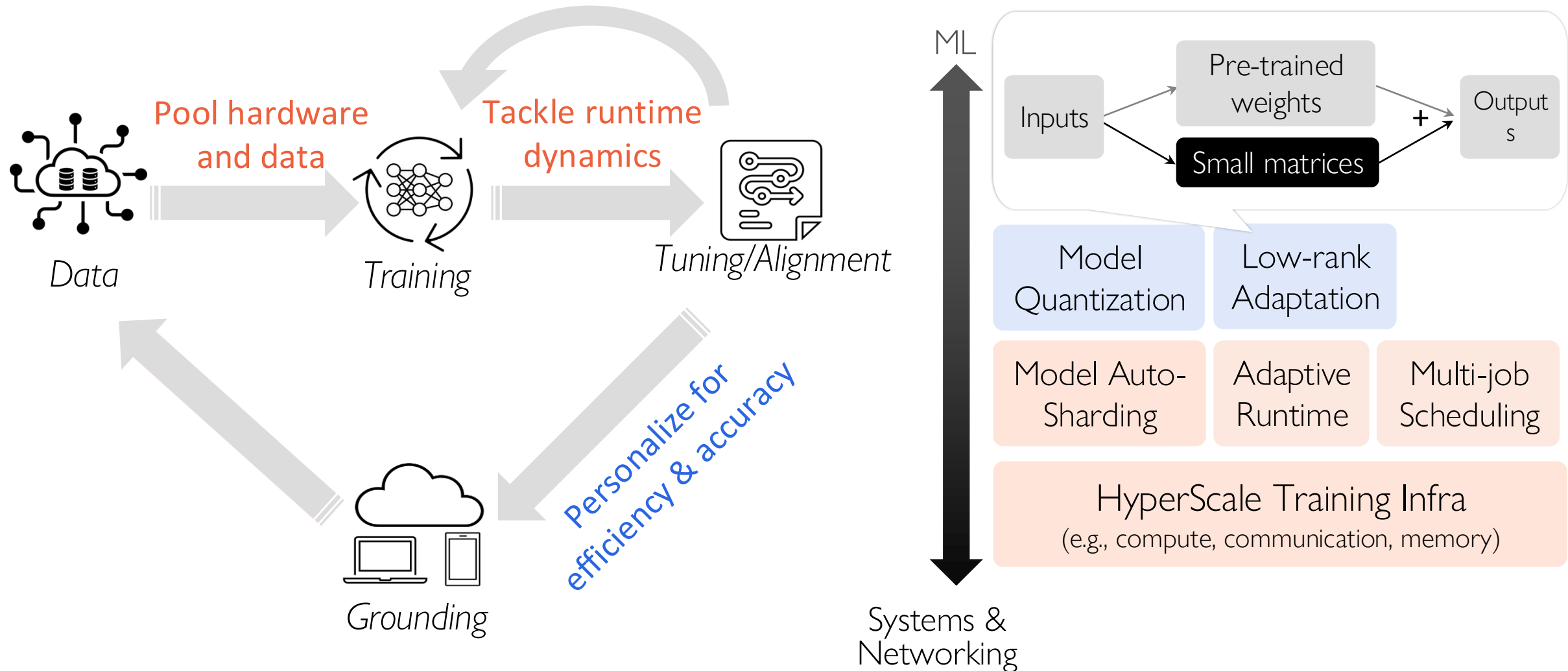
GenAI Lifecycle Demands Great Systems Support



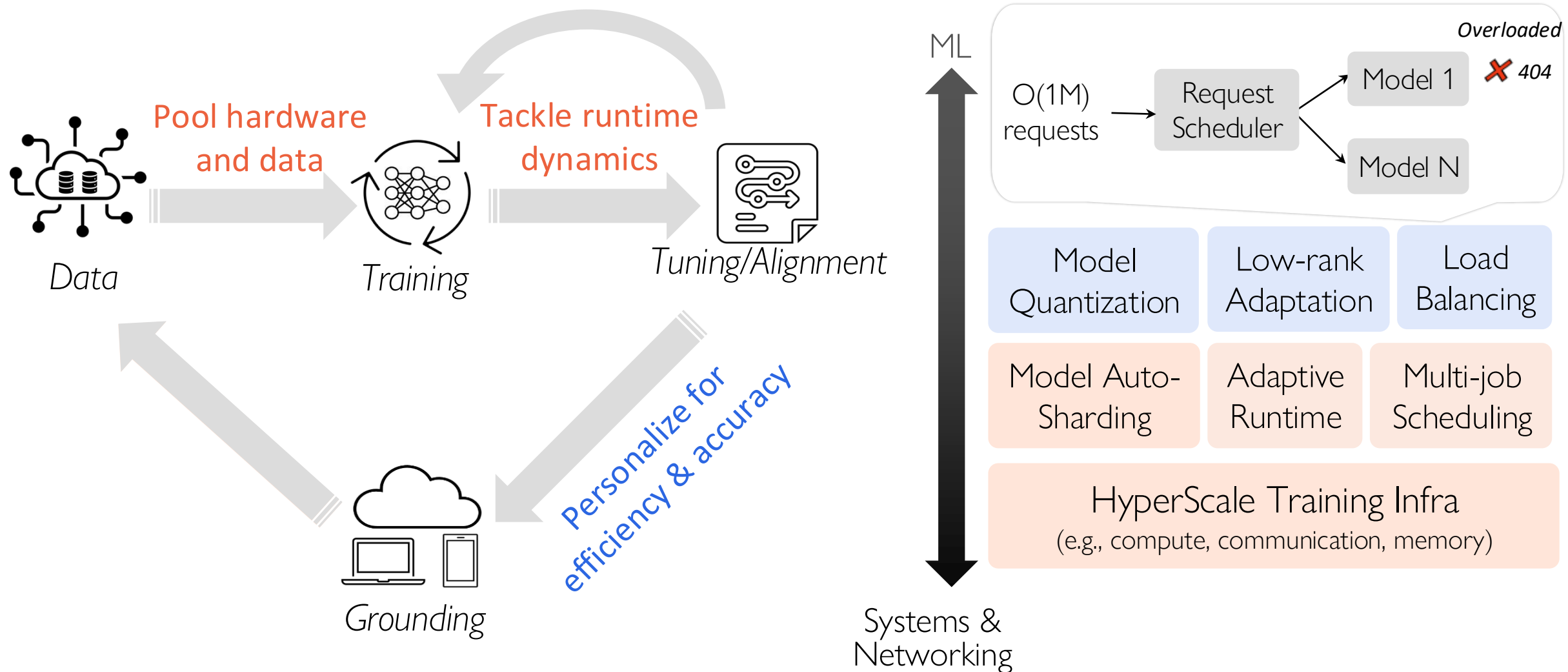
GenAI Lifecycle Demands Great Systems Support



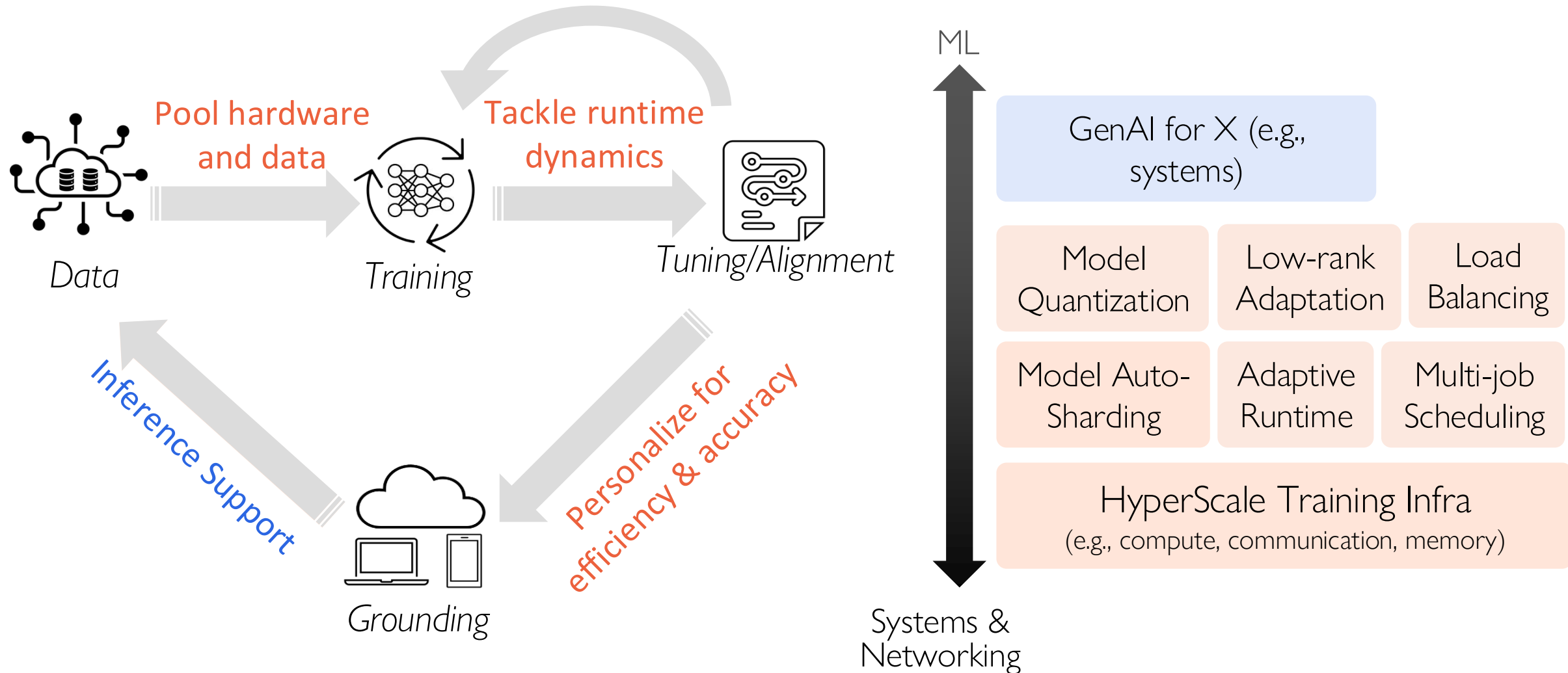
GenAI Lifecycle Demands Great Systems Support



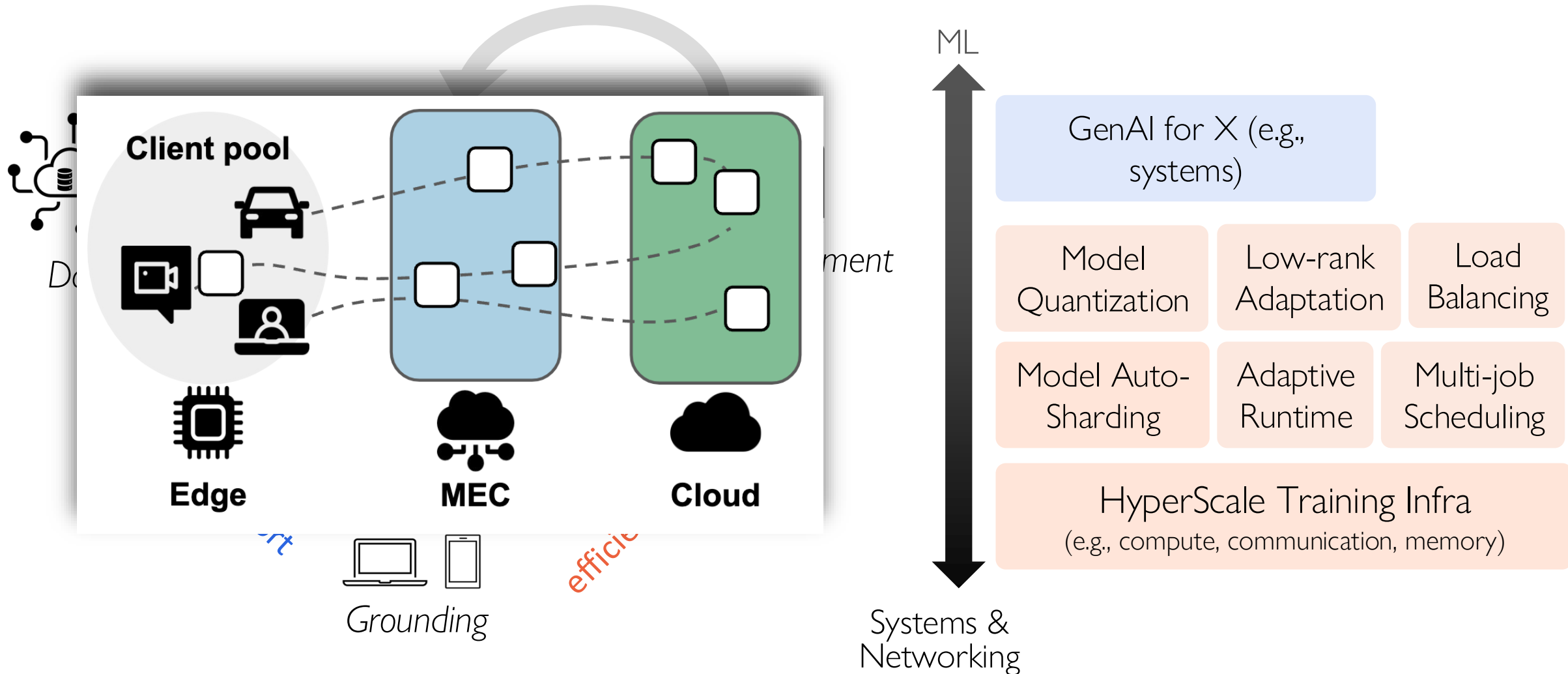
GenAI Lifecycle Demands Great Systems Support



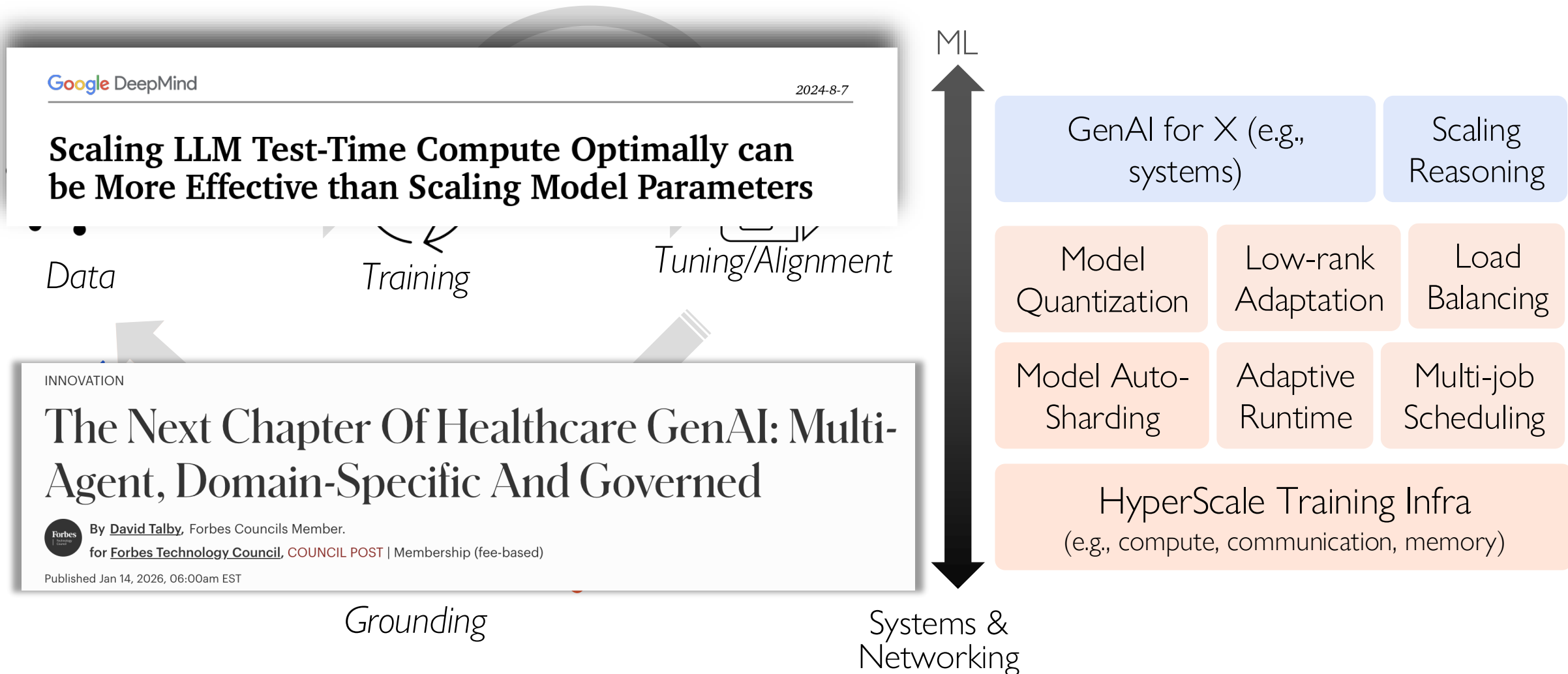
GenAI Lifecycle Demands Great Systems Support



GenAI Lifecycle Demands Great Systems Support



GenAI Lifecycle Demands Great Systems Support



What will you learn in this course?

-- How to **Support Practical ML**

Efficiency: Efficient use of hardware for high scalability and throughput

Effectiveness: High accuracy and fast convergence

Easy to use: Improve development productivity for developers and users

Without ML Systems



ResNet
Transformer

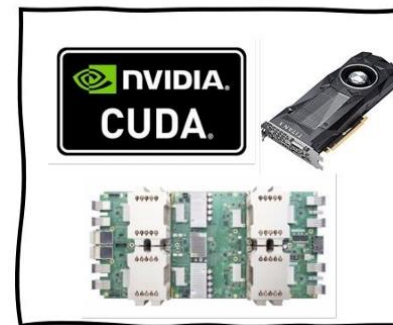
ML Research

44k lines of code

Six months



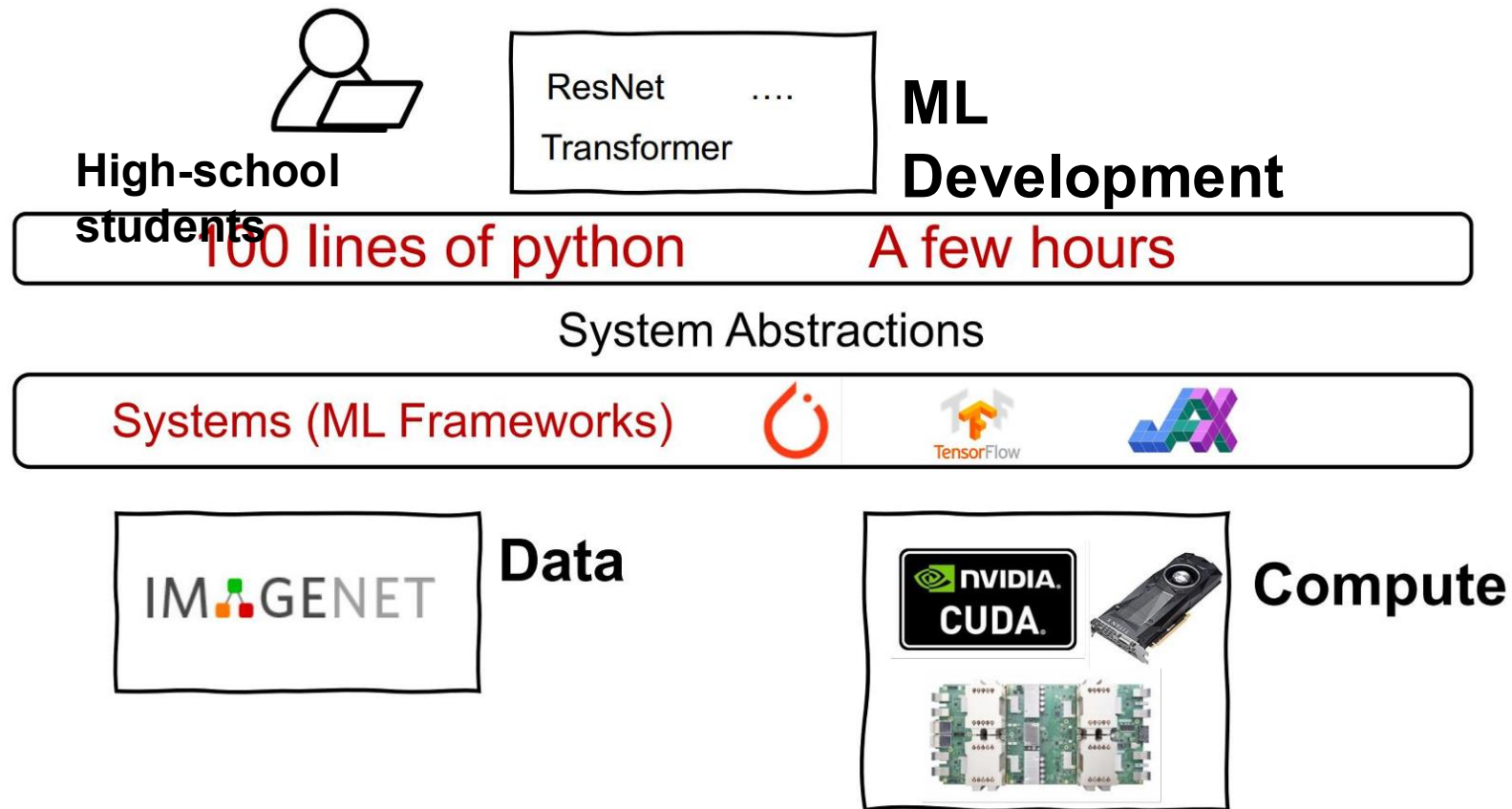
Data



Compute

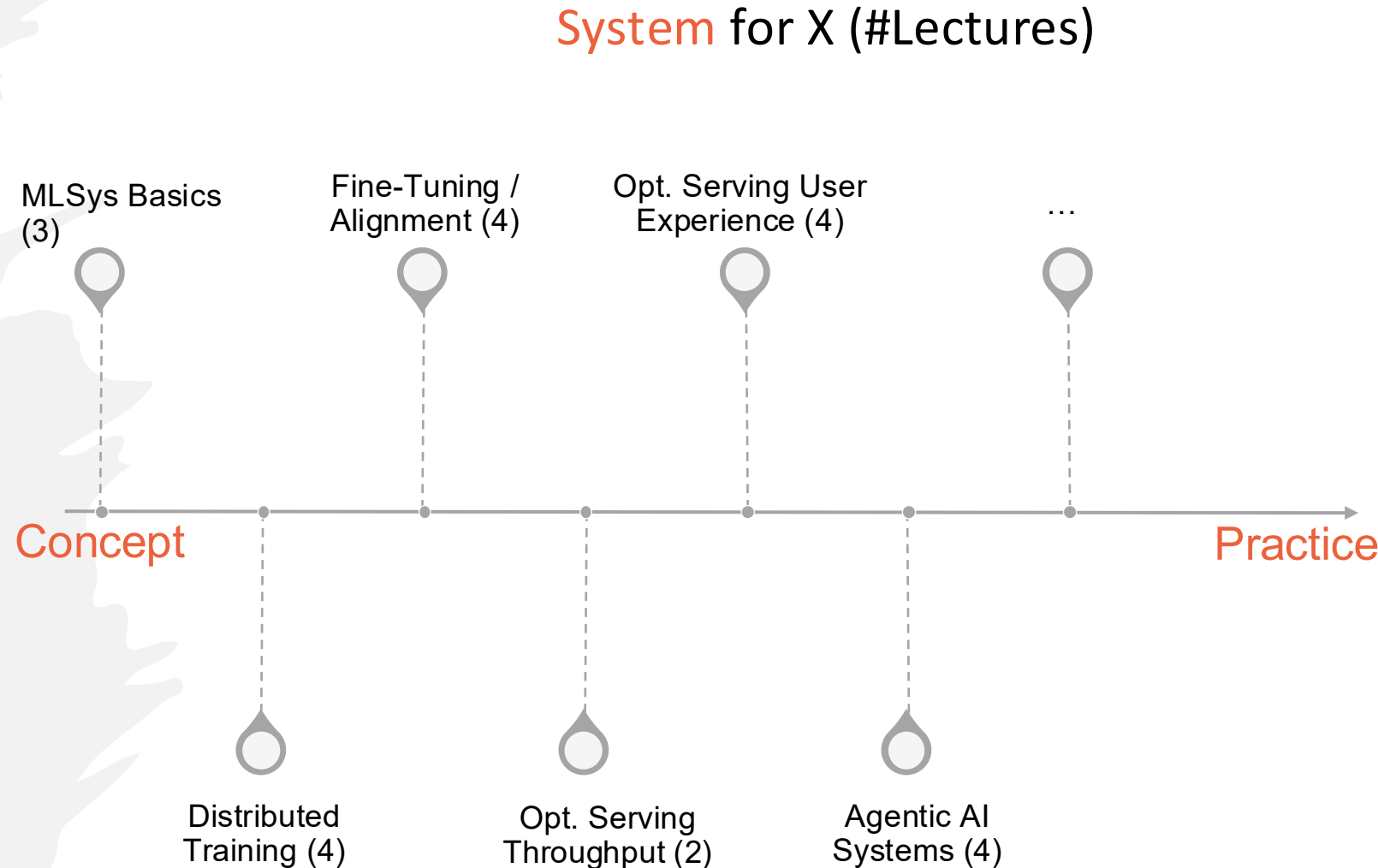
As ML Systems Democratize

AI



What will you learn in this course?

-- How to **Support Practical ML**



What will you learn in this course?

-- How to **Support Practical ML**

Learning Outcomes

After completing this course you should be able to

- Understand the basic functioning of ML Sys
- Articulate and critique latest GenAI Sys
- Be prepared to perform new research

This is a **Seminar-like** Course.
The more **active** the better.

Have Fun!

Agenda

- What will you learn in this course?
 - GenAI, systems, networks, ...
- What will you do in this course?
 - Participation, presentation, paper review, course project, ...

Course Expectations (Grading)

	Weight	Notes
Participation	10%	Up to two absences

While we value teamwork, clarify **each member's** contribution in group **report**.

Grading Scheme (grade is the better of the two)

Grade	Absolute Cutoff ($\geq$)	Relative Bin ($\geq$)
A+	95	Top 5%
$\geq$ A-	88	Top 35%
$\geq$ B-	80	Top 80%
$\geq$ C-	70	Hard to Believe!
$<$ C-		Unbelievable!!

Out of 230+ students, only 5 got $\leq$ B in my previous course. 😊

- Use Canvas to find potential group members

- Group size should be 5

Form Groups ASAP May allow a few smaller groups if/when students drop out

- Submit at <https://forms.gle/xTuaro9vPAptsXir5>
 - By **Feb 1st** the latest, but **right now** is better

Participation & Paper Review

- Up to **two absences** in attendance
 - Write summaries for **14 out of 19 required** papers (feel free to pick 14)

Feb 4 (Diffusion Models)	The Illustrated Stable Diffusion Scalable Diffusion Models with Transformers (Required)	First review due		
	Pre-Training			
Feb 6 (Hybrid Parallelism)	Efficient Large-Scale Language Model Training on GPU Clusters Using Megatron- LM (Required) Alpa: Automating Inter- and Intra-Operator Parallelism for Distributed Deep Learning			

Please submit your PDF summary to Canvas by 11:59 PM **before** lecture day.

Participation & Paper Review

- Each summary (4 **paragraphs**) must follow the template
 - What is the problem and why is it important?
 - The main proposed idea(s). A summary of your understanding of different components of the proposed technique, e.g., the purpose of critical designs
 - What are the **technical** drawbacks/limitations
 - How will you improve them?

Please submit your PDF summary to Canvas by 11:59 PM **before** lecture day.

Participation & Paper Review

- Each summary (4 **paragraphs**) must follow the template
 - What is the problem and why is it important?
 - The main proposed idea(s). A summary of your understanding of different components of the proposed technique, e.g., the purpose of critical designs
 - What are the **technical** drawbacks/limitations
 - How will you improve them?

Grading criteria, each summary has 8 points in total. For each item here, you get:

- 2: The summary item demonstrates a clear understanding of the paper.
- 1: The summary item misses the point of the paper.
- 0: The summary item is missing.

Participation & Paper Review

Not Good Review

- 1. The paper introduces Inter-Galactic Networking, or IGN, which solves the problem of sending messages between galaxies. The authors introduce a novel protocol that can send packets long distances via a distributed array of wormhole actuators.*
- 2. On page 7 the authors misspelled "ambidextrous."*

Better Review

- 1. The simulation was relatively realistic, but didn't measure the total round-trip latency including wormhole setup time for a new destination. So it was hard to know the range of distances at which IGN works better than traditional Inter-Planetary Networking. If most communication is local, IGN would help for only a minority of traffic.*
- 2. The IGN design assumes that the wormhole actuators have already been calibrated to their destinations' coordinates within a certain tolerance. It's not clear how robust this fixed calibration will be given the motion of galaxies while packets are in flight.*
- 3. The theoretical analysis is particularly valuable because their statistical techniques can probably be applied to terrestrial delay-tolerant and underwater networks too.*
- 4.*

Paper Presentation (15%)

- This is a seminar-style course
 - Each group presents **one lecture** covering two papers
- The entire class will be dedicated to the assigned paper(s)
 - Aim for **40-min (e.g., 17+23)** presentation w/o interruption
 - There will be **intermittent discussions**
 - No more than 55 minutes in total (w/ interruption)
- Lead the discussion ([details](#))
 - Go through the paper in details, along with its strengths and weaknesses

Submit slides (in *.ppt format) to [Canvas](#) **48 hrs before** the class for feedbacks

Paper Presentation (15%)

- Please spend a significant amount of time
 - DO NOT read script, DO face to audience
- Graded by instruction team (50%) and your classmates (50%)
 - **Instructor:** based on correctness, depth, clarity, insights, interaction
 - **Peers:** make sure your classmates feel they indeed learn something after listening to your presentation

Submit slides (in *.ppt format) to [Canvas](#) 48 hrs before the class for feedbacks

Panel Discussion (After Normal Lecture)

- The Authors (assigned)
 - ‘Companion’ group that **answers** questions from ‘Reviewers’
- The Reviewers (assigned)
 - ‘Reviewer’ group that **poses** challenging **research** questions
- Rest of the Class & **Instructor**
 - Ask questions directly too

Each group will be assigned to these slots once; 2 discussions per group

Summary of Your Role

Submit Paper
Summary/Slides
(if assigned)

Be Engaged During
Lecture

Right after
teaching

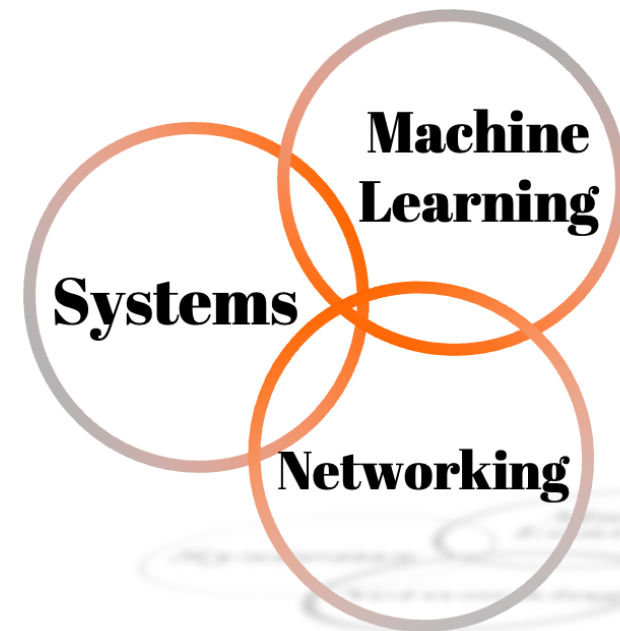
Post-
lecture

One day
before lecture

Attend the Lecture
(Up to 2 absence)

~20 min Pannel Discussion

Systems for ML Projects



Research-Oriented Course Project

- The final project accounts for **50%** of total grades
- What can and cannot be a project?
 - Just surveys are not allowed
 - Measurements of new environments or of existing solutions in new environments are acceptable
 - Proposing new research problems or addressing existing ones is **highly preferred!**
- An ideal project should answer the questions you asked during paper summary and points you cared about in panel discussion

How to Approach it?

1. Find a problem and motivate why this is worth solving
2. Quickly survey background and related work
 - Might require you to go back to the first step
3. Form/update your hypothesis
4. Test your hypothesis
 - Go back to 3 until you are happy
5. Present your findings to the instructor and in writing
 - Discuss known limitations

Draft Proposal (Feb 21st; 5%)

- Two pages ([template](#)), plus references, **must** include
 - *Problem*: What is the problem?
 - *Motivation*: Why is it important?
 - *Solution overview*: Any initial thoughts?
 - *Evaluation plan*: How would you evaluate it?
- Submit as a group, including your group information (*form a group ASAP*)
- Approved by the instructor and agreed upon by you
 - Stop by during OH
 - Forms the basis of expectation

Mid-Semester Report (Mar 27th; 10%)

- Extend your proposal to 4 pages ([template](#)), which must include
 - What is the problem?
 - Why is it important? (Motivating experiments)
 - What are the most related work?
 - What's your hypothesis so far?
 - How are/will you evaluate it?
- Dedicated proposal feedback session (No class on Mar 30, April 3)
 - This is to make sure you are receiving feedback and making progress
 - Feel free to stop by *office hours on any day!*

Final Presentation (Apr 29, May 1 & 6; 15%)

- In-class (group) presentation
 - Please spend a significant amount of time on working your project and making this presentation nice and clear.
- Graded by instruction team (50%) and your classmates (50%)
 - **Instructor:** based on format, correctness, depth, clarity, insights
 - **Peers:** make sure your classmates feel they indeed learn something after listening to your presentation

Final Report (May 15th; 20%)

- Final report (8 pages excluding references by May 15):
 - Should be written like the papers you've read
 - Abstract, Introduction
 - Background/Motivation
 - Design
 - Implementation and Evaluation
 - Conclusion
 - As if you'd submit it to a workshop with ~3 more months of work or to a conference after ~6 more months of work
 - [How to Write a Great Research Paper](#) by Simon Peyton Jones

Project Ideas

- Some potential project ideas (let's chat more)
 - Multi-tier memory, including CXL and RDMA, for long-context LLMs
 - Energy-efficient GenAI serving on end-user devices
 - Systems support for reinforcement learning from human feedback
 - Making chatbots real by allowing them to cut in
 - Repurposing spatiotemporal redundancy in video generation
 - Benchmark important DL workloads to identify important angles
 - ML for systems optimizations (e.g., networking problems)
 - ...
- Any of your crazy ideas!

Thank you!

Date	Milestone	Details
02/01	Form Group	Find 5 like-minded teammates
02/21	Submit Proposal	Submit your proposal to Canvas Receive feedback via in-person OH
	Proposal Feedback	Refine the problem, overview, and form initial hypothesis/idea
03/27	Mid-term Report	4-page report (including motivation numbers)
04/29 05/01 05/06	In-Class Project Presentations	Present your research findings
05/15	Final Report	Submit a report like the papers you read

NO extensions! Delayed submission will receive NO CREDIT